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Hours
[2 Hours]

🎯 Learning Objectives

By the end of this lab, students will:

1. Create a C# Console project in Microsoft Visual Studio.
2. Create and use Lists.
3. Read JSON files.
4. Convert JSON into C# objects (Deserialization).
5. Modify data and save back to JSON.
6. Understand why JSON is important for APIs.
7. Be ready to move to ASP.NET Core Minimal Web API.

1. What is a List?

A List is a collection that stores multiple values in one variable.

2. Create a List

Syntax:

```
List<Type> name = new List<Type>();
```

Example:

```
List<string> names = new List<string>(); or List<int> numbers = new List<int>();
```

Explain: List is the collection. <string> is the type of data.

3. Add Items to a List	4. Display List Items	5. Count Items in a List
`names.Add("Ali");`	`foreach (var n in names)` `{ Console.WriteLine(n); }`	`Console.WriteLine(names.Count);`
6. Remove Items `names.Remove("Sara");`	**7. Search in a List** `bool found = names.Contains("Ali");`	

Write a program that:

1. **Creates a list of 3 student names.**
2. **Displays them.**
3. **Adds one new student from user input.**

List of Objects (Most Important for JSON)

This is the key step before JSON.

Create class:

```
public class Student
{     public string Name { get; set; }    }
```

Now:

```
List<Student> students = new List<Student>();// JSON arrays = List of objects in C#.
students.Add(new Student { Name = "Ali" });
students.Add(new Student { Name = "Sara" });
```

Why Lists Are Important for APIs? When calling an API in ASP.NET Core, the JSON response is often converted into a list in C#.

So this JSON `[ { "Name": "Ali" }, { "Name": "Sara" } ]`

Becomes in C#: **List<Student>**

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Part 1 – Create a Console Project

Students should:

1. Open **Visual Studio**
2. Click **Create a new project**
3. Select **Console App (.NET 8)**
4. Name the project: ☞ StudentJsonLab

Part 2 – Add a JSON File

Create a new file in the project called: ☞ students.json

Paste the following:

```
[
  { "Id": 1, "Name": "Ali", "Age": 21, "Department": "CS" },
  { "Id": 2, "Name": "Sara", "Age": 22, "Department": "IT" },
  { "Id": 3, "Name": "Omar", "Age": 20, "Department": "AI" }
]
```

Then: ☞ Right click file → Properties → Copy to Output Directory → Copy always. This simulates data coming from an API.

Part 3 – Create the Student Class

Create a class:

```
public class Student
{   public int Id { get; set; }
    public string Name { get; set; }
    public int Age { get; set; }
    public string Department { get; set; }        }
```

📌 JSON keys must match class properties.

Part 4 – Read JSON and Display Data

Students will:

1. Read the JSON file.
2. Convert JSON into objects.
3. Display all students.

```
using System.Text.Json;

string json = File.ReadAllText("students.json");

List<Student> students =    JsonSerializer.Deserialize<List<Student>>(json);

foreach (var s in students)

{    Console.WriteLine($"{s.Id} - {s.Name} - {s.Department}");    }

Console.WriteLine(students.Any(s => s.Name == "Ali"));
```

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Part 5 – Add a New Student and Save JSON

Students should:

1. Ask user to enter a new student.
2. Add it to the list.
3. Save the updated list back to JSON.

```
Console.Write("Enter Name: ");

string name = Console.ReadLine();

Console.Write("Enter Age: ");

int age = int.Parse(Console.ReadLine());

Console.Write("Enter Department: ");

string dept = Console.ReadLine();

Student newStudent = new Student

{  Id = students.Max(s => s.Id) + 1, // s => s.Id → is a lambda expression. It means:

"From each student s, take the value of Id."

    Name = name,
    Age = age,
    Department = dept
};

students.Add(newStudent);
string newJson = JsonSerializer.Serialize(students,
    new JsonSerializerOptions { WriteIndented = true });
File.WriteAllText("students.json", newJson);
Console.WriteLine("Student added successfully!");
```

🎯 **Task**

- Add a feature (Grade) to the student object:
- Search for a student by ID.
- Update a student.
- 📝 **Quick Quiz (5 min)**

1. JSON is mainly used for: A. Designing UI B. Storing and transferring data C. Compiling code D. Styling web pages	**3. What happens if JSON property names do not match class names?** A. Program crashes B. It still works C. Data is ignored or null D. Faster execution
2. Which method converts JSON into C# objects? A. Serialize B. Deserialize C. Convert D. Parse	**4. Why do Web APIs use JSON?** A. Easy to read B. Lightweight C. Language independent D. All of the above